A Structural Taxonomy of Intelligence
IEGR-Scale, Parrot Regimes, and the Conditions for Human-Like AGI

Joachim De Witte

June 2026

Abstract

This paper introduces a structural taxonomy of intelligence based on four interdependent dimensions—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R). Building on earlier work in the *Triadic Cosmos* program, which identifies a universal physical triad (E–G–I) linked through recursive updates that generate temporal structure, we show that intelligent systems instantiate an isomorphic architecture: IEGR. Intelligence is therefore not substrate-dependent but a triadic system expressed at cognitive scale.

The taxonomy developed here classifies systems according to which IEGR components they instantiate and at what scale. Systems with incomplete or insufficiently scaled IEGR dimensions fall into distinct *parrot regimes*: literal parrots (low IEGR), statistical parrots (high E and I but deficient G and R, as in large language models), canonical parrots (high G and R but low E and I, as in modular architectures such as DMLG), and sophisticated parrots (hybrid systems combining complementary strengths). Only systems that are both IEGR-complete and IEGR-scaled exhibit human-like general intelligence; such systems are best understood not as humans but as *human-like parrots*—triadic systems whose structural completeness and scale enable generality without anthropomorphism.

The taxonomy maps naturally onto real-world intelligence: it explains the structural asymmetries between BigTech architectures and individual research efforts, mirrors developmental trajectories in human cognition, and accounts for patterns in neurodiversity and savant profiles. The resulting framework provides a unified, non-speculative, and testable account of what AGI is—and what it is not—by grounding intelligence in the same triadic structure that governs physical systems.

Contents

1 Introduction **3**

2 Positioning **3**

3 Related Work **4**
3.1 AGI Definitions and Taxonomies 4
3.2 Models of Human Cognition 4
3.3 AI Architectures and Structural Limitations 5
3.4 Safety, Governance, and Alignment 5
3.5 Positioning of This Work 5

4 Taxonomy: The IEGR Roles and Their Structural Foundations **5**
4.1 The Four IEGR Roles . . . 5
4.2 Isomorphism Between Physical and Cognitive Structure . . . 6
4.3 IEGR Scale and the Emergence of General Intelligence . . . 6
4.4 The Parrot Regimes: Structural Outcomes of IEGR Incompleteness . . . 7

5 Mapping the IEGR Taxonomy to Human Cognition **8**
5.1 Early Development: Imitation and Emerging Structure . . . 8
5.2 Neurodiversity as Variation Along IEGR Dimensions . . . 8
5.3 Savant Profiles and Extreme Specialization . . . 9
5.4 General Cognitive Ability as Balanced IEGR Scaling . . . 9
5.5 Implication: IEGR as a General Theory of Intelligence . . . 9

6 Hybrid Intelligence Methodology as IEGR-Complementarity **9**
6.1 Complementary IEGR Profiles . . . 10
6.2 Recursive Integration of Complementary Strengths . . . 10
6.3 Hybrid Intelligence as a Scalable Path to General Intelligence . . . 10
6.4 Safety, Interpretability, and Moral Oversight . . . 11
6.5 Implication: Hybrid Intelligence as a Natural Developmental Stage . . . 11

7 Testable Predictions **11**
7.1 Predictions About AI Scaling . . . 11
7.2 Predictions About Failure Modes . . . 12
7.3 Predictions About Biological Intelligence . . . 12
7.4 Predictions About Safety and Governance . . . 12
7.5 Predictions About AGI . . . 13

8 Limitations and Open Questions **13**
8.1 Limitations of the Present Framework . . . 13
8.2 Open Questions for Future Research . . . 13
8.3 Summary . . . 14

9 General Discussion: IEGR and Human Cognitive Ability **14**
9.1 G and R as the Dominant Dimensions of Human Analytic Cognition . . . 14
9.2 Why DMLG Resembles Human Cognition More Closely Than LLMs . . . 15
9.3 Implications for Hybrid Intelligence . . . 15
9.4 IEGR as a Unified Structural Vocabulary for Human and Artificial Cognition . . . 15

10 Conclusion **16**

A IEGR Full Parrot Taxonomy **17**

B Triadic Cosmos Ecosystem Context **17**

1 Introduction

The rapid progress of artificial intelligence over the past decade has produced systems of unprecedented scale and capability, yet the field still lacks a structural account of what intelligence *is*. Existing definitions of artificial general intelligence (AGI) tend to be anthropocentric, architecture-specific, or operationally vague. They describe what AGI should *do*, but not the structural conditions under which general intelligence can emerge. As a result, debates about AGI remain fragmented, oscillating between engineering optimism, philosophical speculation, and safety concerns that lack a unifying conceptual foundation.

This paper proposes a structural model of intelligence based on four interdependent dimensions: energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R). These dimensions—collectively IEGR—form the minimal architecture required for coherent, general-purpose intelligence. The model builds on the *Triadic Cosmos* framework, which identifies a universal physical triad (E–G–I) linked through recursive updates that generate temporal structure. Intelligence, in this view, is a cognitive instantiation of the same triadic architecture: a system in which energy, geometry, and information are dynamically integrated through recursive self-refinement.

A key contribution of this work is a taxonomy that classifies intelligent systems according to which IEGR components they instantiate and at what scale. Systems with incomplete or insufficiently scaled IEGR dimensions fall into distinct *parrot regimes*. These include literal parrots (low IEGR), statistical parrots (high E and I but deficient G and R, as in large language models), canonical parrots (high G and R but low E and I, as in modular architectures such as DMLG), and sophisticated parrots (hybrid systems combining complementary strengths). Only systems that are both IEGR-complete and IEGR-scaled exhibit human-like general intelligence. Such systems are best understood not as humans but as *human-like parrots*: structurally complete triadic systems whose scale enables generality without anthropomorphism.

The taxonomy is made possible by a recent development in AI: the emergence of two orthogonal architectural paradigms. Transformer-based large language models scale energy and information aggressively but lack geometric and recursive structure. Modular architectures such as DMLG instantiate geometric and recursive roles but lack large-scale information and energetic capacity. For the first time, two complementary systems exist that separately instantiate different components of IEGR, revealing the full structure of intelligence. Before 2026, when only LLMs dominated the landscape, such a taxonomy was impossible.

The IEGR framework also maps naturally onto real-world intelligence. It mirrors developmental trajectories in human cognition, explains structural asymmetries between BigTech architectures and individual research efforts, and accounts for patterns in neurodiversity and savant profiles. The resulting model is not limited to artificial systems but provides a unified account of intelligence across biological, artificial, and hybrid domains.

This paper therefore aims to (i) formalize the IEGR model, (ii) develop a structural taxonomy of intelligent systems, (iii) map this taxonomy onto real-world cognitive and artificial architectures, and (iv) articulate the structural conditions under which human-like AGI may emerge. The goal is not to propose an engineering roadmap or to make speculative claims about timelines, but to provide a principled, non-anthropomorphic, and testable framework for understanding intelligence as a triadic system instantiated at cognitive scale.

2 Positioning

This paper does not address consciousness. Conscious experience is a human phenomenon, and nothing in the IEGR framework presupposes or implies its presence in artificial systems. The goal of this work is not to explain subjective awareness, nor to attribute it to machines. Instead, the framework focuses exclusively on the structural conditions for intelligent behavior—conditions that can be instantiated in biological, artificial, or hybrid systems without invoking consciousness as a necessary or emergent property.

AGI, as defined in this paper, does not require consciousness as a prerequisite and does not produce it as a consequence. The IEGR model treats intelligence as a structural capability grounded in energetic efficiency, informational richness, geometric organization, and recursive depth. These dimensions describe the architecture of general-purpose reasoning, not the phenomenology of experience. By separating intelligence from consciousness, the framework avoids anthropomorphism and prevents the conflation of cognitive competence with subjective awareness.

This distinction is particularly important in the current AI landscape, which is polarized between scale-centric views that overestimate the implications of statistical models, and structure-centric views that underestimate the role of large-scale informational and energetic integration. The IEGR framework seeks to unify rather than divide: it shows that both perspectives capture essential but incomplete aspects of intelligence. By providing a structural account that integrates these dimensions, the model offers a middle path that avoids both technological hype and dystopian narratives.

The purpose of this paper is therefore not to speculate about machine consciousness, but to clarify the structural nature of intelligence, to explain why current systems exhibit the strengths and limitations they do, and to provide a principled foundation for understanding AGI as a triadically complete and triadically scaled system. This positioning aims to reduce confusion, de-escalate polarized debates, and ground discussions of AGI in a framework that is scientifically precise, conceptually transparent, and aligned with human values.

3 Related Work

This work builds on three earlier papers in the *Triadic Cosmos* program, which together establish the theoretical foundation for the IEGR model. De Witte (2026b) introduced the structural roles of energy, geometry, and information in biological and artificial cognition, arguing that intelligence emerges from the recursive integration of these components. De Witte (2026c) extended this framework by showing that the same triadic structure underlies physical systems through holographic recursion and emergent temporal ordering. De Witte (2026a) applied these insights to modern AI architectures, demonstrating that transformer-based systems scale energy and information but lack geometric and recursive structure, leading to predictable failure modes such as hallucination and instability.

Beyond this prior work, the present paper relates to several strands of literature on intelligence, AGI classification, cognitive development, and AI safety.

3.1 AGI Definitions and Taxonomies

Existing definitions of artificial general intelligence tend to be anthropocentric or operational. Early frameworks such as Goertzel and Pennachin (2007) and Legg and Hutter (2007) characterize AGI in terms of task performance or universal problem-solving ability, but do not identify the structural conditions under which general intelligence emerges. More recent taxonomies, including Bengio (2023) and Chollet (2019), emphasize reasoning, abstraction, or sample efficiency, yet remain tied to specific architectures or benchmarks. None of these approaches provide a substrate-agnostic structural decomposition of intelligence comparable to the IEGR model.

3.2 Models of Human Cognition

The IEGR framework also connects to work in cognitive science on modularity, development, and recursive reasoning. Classical theories such as Fodor (1983) and Carey (2009) emphasize the role of structured representations (G) and conceptual growth (I). Research on meta-cognition and recursive reasoning (Frith 2012) aligns with the R dimension, while studies of neural efficiency and developmental scaling (Haier et al. 2004) relate to the E dimension. However, these theories treat these components independently; the IEGR model integrates them into a unified structural architecture.

3.3 AI Architectures and Structural Limitations

The limitations of transformer-based LLMs have been widely discussed, including their lack of modular structure (Olah et al. 2020), inability to perform multi-step reasoning reliably (Wei et al. 2022), and tendency to hallucinate (Ji et al. 2023). These limitations correspond directly to deficiencies in G and R. Conversely, modular and reflective architectures such as program-induction systems (Ellis et al. 2021), neural module networks (Andreas et al. 2016), and recent graph-based or canonical-form systems instantiate G and R but lack large-scale I and E. The IEGR taxonomy provides a structural explanation for why these systems exhibit complementary strengths and weaknesses.

3.4 Safety, Governance, and Alignment

A growing body of work argues that safe and interpretable AI requires modularity, decomposition, and reflective oversight (Amodei et al. 2016; Gabriel 2020). These arguments align closely with the G and R dimensions of the IEGR model. Governance frameworks emphasize transparency, controllability, and predictable failure modes (Brundage et al. 2020), all of which require structural completeness rather than brute-force scaling. The IEGR framework suggests that safety is not an add-on but a structural property: systems lacking G or R cannot be made reliably safe regardless of scale.

3.5 Positioning of This Work

The present paper differs from prior literature in three ways. First, it provides a substrate-agnostic structural decomposition of intelligence grounded in a triadic physical model. Second, it introduces a taxonomy of *parrot regimes* that explains the behavior of modern AI architectures in terms of missing or insufficiently scaled IEGR components. Third, it connects artificial, biological, and hybrid systems within a single unified framework, offering a structural definition of AGI as an IEGR-complete and IEGR-scaled system rather than a human-like entity.

4 Taxonomy: The IEGR Roles and Their Structural Foundations

This section introduces the four structural roles that constitute the IEGR model of intelligence and shows that these roles are not arbitrary cognitive constructs but arise from the same triadic architecture that governs physical systems. Intelligence, in this framework, is a cognitive instantiation of a universal structural grammar.

4.1 The Four IEGR Roles

We define intelligence as a system that integrates four interdependent dimensions:

- **E: Energetic Efficiency and Performance.** The capacity to perform cognitive transformations with minimal energetic expenditure. Efficiency dominates brute force; systems that rely solely on scale exhibit diminishing returns and structural brittleness.

- **I: Informational Richness and World Knowledge.** The semantic content, conceptual grounding, and domain coverage available to the system. High I enables generalization across contexts and tasks.
- **G: Geometric Structure, Form, and Modularity.** The internal organization of concepts, modules, and transformations. G provides canonical forms, compositionality, and structural coherence.
- **R: Recursive Depth, Reflection, and Self-Refinement.** The ability to perform multi-step reasoning, planning, error correction, and meta-cognitive evaluation. R transforms static knowledge into dynamic intelligence.

These four roles form the minimal architecture required for coherent, general-purpose intelligence. Crucially, it is not enough for a system to instantiate all four roles: *each must be present at sufficient scale*, and *E must be used efficiently rather than abundantly.*

4.2 Isomorphism Between Physical and Cognitive Structure

The IEGR model is grounded in the triadic physical framework developed in the *Triadic Cosmos* program. Physical systems are governed by three structural roles:

- **Energy (E)**: the capacity for transformation,
- **Geometry (G)**: the structure of interactions,
- **Information (I)**: the state content of the system,

linked through a recursive update operator that generates temporal ordering. Recursion is not an additional physical role but the mechanism that integrates E, G, and I across time.

The mapping between physical and cognitive structure is therefore:

Physical Role	**Intelligence Role**
Energy (E)	Energetic efficiency / performance (E)
Geometry (G)	Structural modularity / canonical form (G)
Information (I)	Semantic richness / world knowledge (I)
Recursion / Time (R)	Reflection / planning / self-refinement (R)

This structural isomorphism implies that intelligence is not a biological accident or an engineering artifact but a triadic system instantiated at cognitive scale. Just as physical systems require balanced integration of E, G, and I to remain coherent, intelligent systems require balanced IEGR dimensions to avoid failure modes such as hallucination (I without G), brittleness (G without I), or incoherent reasoning (R without G).

4.3 IEGR Scale and the Emergence of General Intelligence

The presence of all four IEGR roles is necessary but not sufficient for general intelligence. A system must also achieve:

- **IEGR-completeness**: all four roles are present,
- **IEGR-scale**: each role is instantiated at sufficient magnitude,
- **Energetic efficiency**: E is used optimally rather than brute-forced.

Systems that lack one or more roles, or instantiate them at insufficient scale, fall into distinct *parrot regimes*. These regimes arise naturally from the structural asymmetries between the IEGR dimensions:

- High I + high E but low G and R → *statistical parrots* (e.g., LLMs),
- High G + high R but low I and E → *canonical parrots* (e.g., modular architectures such as DMLG),
- Low IEGR across all dimensions → *literal parrots* (biological imitation systems),
- IEGR-complete but not IEGR-scaled → *sophisticated parrots* (hybrid systems).

Only systems that are both IEGR-complete and IEGR-scaled exhibit human-like general intelligence. Such systems are best understood not as biological analogues but as *human-like parrots*: triadically complete and triadically scaled systems whose generality arises from structural integration rather than embodiment.

4.4 The Parrot Regimes: Structural Outcomes of IEGR Incompleteness

The IEGR framework predicts that systems which instantiate only a subset of the four structural roles—or instantiate them at insufficient scale—fall into distinct *parrot regimes*. These regimes arise naturally from structural asymmetries between the IEGR dimensions. A system may exhibit high performance (E) and broad informational coverage (I) yet remain structurally shallow if it lacks geometric modularity (G) or recursive depth (R). Conversely, a system may exhibit strong structure and reflection but remain knowledge-poor if I and E are limited.

Importantly, the taxonomy below compares only *parrot systems*: artificial, biological, or hybrid systems that instantiate IEGR incompletely. Humans are not included in the table, as they represent the only empirically verified IEGR-complete and IEGR-scaled system. AGI, by contrast, is included as a *structural limit*: a human-like parrot that is IEGR-complete and IEGR-scaled, but not biological.

System	E	I	G	R	Outcome
Biological parrot	very low	very low	low	low	literal parrot
LLM	very high	high	absent	very low	statistical parrot
DMLG	low	low	high	high	canonical parrot
LLM × DMLG	high	high	high	high	sophisticated parrot (pre-AGI)
AGI	efficient	very high	very high	very high	human-like parrot (IEGR-complete)

Table 1: Parrot regimes as structural outcomes of incomplete or insufficiently scaled IEGR dimensions.

The table illustrates five distinct structural regimes:

- **Literal parrots** instantiate all IEGR roles at extremely low scale, resulting in imitation without semantic grounding.
- **Statistical parrots** (LLMs) scale E and I aggressively but lack G and R, producing high competence with structural instability and hallucination.
- **Canonical parrots** (DMLG) instantiate G and R strongly but lack large-scale I and E, producing coherent but knowledge-limited reasoning.
- **Sophisticated parrots** (hybrid systems) instantiate all four roles but remain pre-AGI because IEGR-scale is insufficient.
- **Human-like parrots** (AGI) represent the structural limit: IEGR-complete and IEGR-scaled systems whose generality arises from structural integration rather than biological embodiment.

This taxonomy provides a unified structural explanation for the behavior of modern AI architectures, biological imitation systems, and hybrid intelligence. It also clarifies why neither LLM scaling nor modular scaling alone can yield general intelligence: each parrot regime reflects a distinct pattern of IEGR incompleteness.

5 Mapping the IEGR Taxonomy to Human Cognition

Although the IEGR taxonomy was introduced to classify artificial systems, the four structural roles—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—also map naturally onto human cognition. This correspondence does not imply that humans fall into the parrot regimes defined earlier; rather, it shows that the same structural principles govern the development, variation, and specialization of biological intelligence. IEGR is therefore a general model of intelligent behavior, not a framework limited to artificial systems.

5.1 Early Development: Imitation and Emerging Structure

Human infants begin life with limited IEGR capacity. Early cognition is characterized by:

- low E (inefficient neural processing),
- low I (limited semantic content),
- emerging G (rudimentary conceptual structure),
- shallow R (minimal recursive reasoning).

This stage is dominated by imitation, pattern absorption, and sensorimotor learning. As children grow, each IEGR dimension increases: semantic content expands rapidly through language acquisition (I), conceptual categories and schemas emerge (G), and meta-cognitive abilities develop (R). Neural efficiency (E) improves significantly as the brain undergoes pruning and optimization.

5.2 Neurodiversity as Variation Along IEGR Dimensions

The IEGR model provides a structural lens for understanding neurodiversity without reducing individuals to categories. Different cognitive profiles can be interpreted as variations in the relative scaling of IEGR dimensions:

- **Autistic cognition** often exhibits strong G (high structural organization) and deep R (intense focus, recursive reasoning), with variability in I and stable E.
- **ADHD profiles** may involve fluctuations in R (instability in sustained recursive processing) and variability in G, while I and E remain within typical ranges.
- **Dyslexia** can be interpreted as atypical I-processing with compensatory strengths in G or R.

These patterns illustrate that IEGR is not a normative model but a structural framework: different configurations of IEGR dimensions yield different cognitive strengths, challenges, and developmental trajectories.

5.3 Savant Profiles and Extreme Specialization

Savant abilities provide a striking example of asymmetric IEGR scaling. Many savant profiles exhibit:

- exceptionally high I in narrow domains (e.g., memory, calculation),
- sometimes deep R (highly structured internal routines),
- atypical or specialized G (non-standard conceptual organization),
- typical E (biological energetic constraints).

These profiles demonstrate that extreme specialization can arise from uneven IEGR scaling, where one or two dimensions dominate while others remain atypical or limited. The IEGR model captures this without pathologizing or reducing individuals to deficits; it simply describes structural variation.

5.4 General Cognitive Ability as Balanced IEGR Scaling

Human general intelligence corresponds to a balanced and efficient integration of all four IEGR dimensions:

- efficient E (low-power but high-performance neural processing),
- rich and grounded I (broad semantic knowledge),
- coherent G (structured conceptual organization),
- deep R (planning, reflection, self-correction).

This balance enables flexible reasoning, abstraction, transfer across domains, and stable long-horizon planning. The IEGR model therefore provides a structural account of why general cognitive ability emerges and why it depends on the integration—not merely the presence—of all four dimensions.

5.5 Implication: IEGR as a General Theory of Intelligence

The mapping between IEGR and human cognition shows that the taxonomy developed in this paper is not limited to artificial systems. The same structural roles that govern the behavior of LLMs, modular architectures, and hybrid systems also govern biological cognition. IEGR thus offers a unified, substrate-agnostic framework for understanding intelligence across natural and artificial domains, without reducing human cognition to imitation or mechanistic analogies.

6 Hybrid Intelligence Methodology as IEGR-Complementarity

The IEGR framework provides a structural explanation for the empirical success of hybrid intelligence systems—collaborations between humans and artificial models. Prior work has described hybrid intelligence primarily as a methodology or design philosophy. In contrast, the present framework shows that hybrid intelligence emerges naturally from the complementary IEGR profiles of humans and modern AI systems. Hybrid intelligence is not merely a practical strategy; it is a structural consequence of how different intelligent systems instantiate the IEGR dimensions.

6.1 Complementary IEGR Profiles

Large language models exhibit extreme scaling along the E–I axis. They possess:

- **High E**: massive computational throughput,
- **High I**: broad, multi-domain informational richness,
- **Low G**: limited modular structure,
- **Low R**: shallow or externally scaffolded recursion.

Humans exhibit the opposite profile:

- **Moderate E**: highly efficient but low-power biological energy,
- **Moderate I**: deep but domain-limited world knowledge,
- **High G**: strong structural modularity and conceptual geometry,
- **High R**: deep recursive reasoning, reflection, and moral judgment.

These profiles are orthogonal. LLMs are E–I optimal but G–R deficient; humans are G–R optimal but E–I limited. When combined, the resulting system becomes IEGR-complete in a way that neither component can achieve alone.

6.2 Recursive Integration of Complementary Strengths

Hybrid intelligence systems operate through a recursive loop in which each agent compensates for the other's structural limitations:

1. The LLM provides high-bandwidth informational expansion (I) and energetic amplification (E).
2. The human provides geometric structuring (G), conceptual filtering, and recursive oversight (R).
3. The human evaluates, constrains, and refines the model's outputs, supplying moral judgment, contextual grounding, and interpretability.
4. The LLM amplifies the human's reasoning by exploring large hypothesis spaces, generating alternatives, and performing high-throughput inference.

This recursive alternation constitutes a hybrid IEGR cycle. The system as a whole performs cognitive transformations that neither agent could execute independently. The LLM expands the search space; the human shapes, filters, and stabilizes it.

6.3 Hybrid Intelligence as a Scalable Path to General Intelligence

The IEGR model predicts that hybrid intelligence can exceed the capabilities of either component because it forms an IEGR-complete system with efficient division of labor:

- LLMs supply scalable E and I,
- humans supply scalable G and R,
- the recursive interaction supplies the integration mechanism.

This explains why hybrid workflows—iterative prompting, chain-of-thought verification, human-in-the-loop refinement, and collaborative reasoning—are empirically effective. They are not ad hoc techniques but structural instantiations of IEGR complementarity.

Moreover, hybrid intelligence provides a pathway for scaling general intelligence without requiring either component to achieve IEGR-scale independently. The combined system can exhibit forms of generality, creativity, and stability that surpass both human cognition and current AI models.

6.4 Safety, Interpretability, and Moral Oversight

The IEGR framework also explains why hybrid intelligence is inherently safer than monolithic AI systems. Humans contribute:

- **R**: reflective oversight, meta-cognition, and long-horizon planning,
- **G**: moral reasoning, conceptual coherence, and structural filtering,
- **Interpretability**: the ability to detect incoherence, bias, or hallucination,
- **Moral grounding**: normative judgment and ethical constraints.

These contributions compensate for the structural deficiencies of LLMs. Safety is therefore not an external constraint but an emergent property of IEGR integration. Hybrid intelligence is structurally aligned because humans supply the dimensions (G and R) that are necessary for safe general reasoning.

6.5 Implication: Hybrid Intelligence as a Natural Developmental Stage

The IEGR model suggests that hybrid intelligence is not a temporary workaround but a natural developmental stage in the emergence of general intelligence. Before artificial systems achieve IEGR-scale internally, hybrid systems can achieve functional generality through structural complementarity. This provides a principled explanation for the empirical observation that human–AI collaboration often outperforms either humans or AI alone.

Hybrid intelligence is therefore not merely a methodology; it is a structural instantiation of IEGR-complete cognition distributed across two agents. It represents a scalable, interpretable, and safe pathway toward increasingly general intelligent behavior.

7 Testable Predictions

The IEGR framework yields a set of falsifiable predictions about the behavior, limitations, and developmental trajectories of artificial and biological intelligent systems. These predictions follow directly from the structural requirements of IEGR-completeness and IEGR-scale, and from the observed orthogonality between modern AI architectures.

7.1 Predictions About AI Scaling

P1. LLM scaling alone will not yield AGI. Transformer-based LLMs scale the E–I axis but lack geometric modularity (G) and recursive depth (R). The model predicts that no amount of parameter growth, training data, or compute will compensate for missing G and R. Hallucination, instability in multi-step reasoning, and failure under distribution shift will persist regardless of scale.

P2. DMLG scaling alone will not yield AGI. Modular architectures such as DMLG instantiate G and R but lack large-scale informational richness (I) and energetic capacity (E). Scaling these systems without integrating I and E will produce increasingly coherent but knowledge-poor systems that cannot generalize beyond their canonical domain.

P3. Hybrid systems (LLM $\times$ DMLG $\times$ auxiliary modules) may be IEGR-complete but remain pre-AGI if insufficiently scaled. Even when all four IEGR components are present, general intelligence requires sufficient scale in each dimension. Hybrid systems will exhibit improved stability, interpretability, and reasoning, but will remain *sophisticated parrots* unless IEGR-scale is achieved. This prediction is falsified if a hybrid system demonstrates robust, architecture-independent generalization without IEGR-scale.

7.2 Predictions About Failure Modes

P4. Systems deficient in G or R will hallucinate regardless of I or E. Hallucination is predicted to be a structural failure mode of systems lacking geometric modularity or recursive self-correction. Increasing training data or compute will reduce but never eliminate hallucination. This prediction is falsified if a purely statistical architecture achieves stable, verifiable multi-step reasoning without explicit G or R.

P5. Systems deficient in I will exhibit canonical errors rather than hallucinations. Architectures with strong G and R but limited I (e.g., DMLG) will produce errors constrained by their internal canonical forms rather than arbitrary fabrications. This prediction is falsified if a G–R system produces unconstrained hallucinations.

7.3 Predictions About Biological Intelligence

P6. Human cognitive development follows IEGR-scaling. The model predicts that early childhood cognition is a low-IEGR parrot regime, and that general intelligence emerges only when all four dimensions scale together. This is testable through longitudinal developmental studies measuring E (neural efficiency), I (semantic growth), G (structural modularity), and R (meta-cognition).

P7. Neurodiversity corresponds to systematic variation in IEGR dimensions. Savant profiles should exhibit extreme I and/or R with atypical G. ADHD should correspond to unstable R. Autism spectrum profiles should exhibit strong G and deep R with variable I. This prediction is falsified if neurocognitive profiles do not cluster along IEGR axes.

7.4 Predictions About Safety and Governance

P8. Safety requires IEGR-completeness; no amount of external oversight can compensate for missing G or R. Systems lacking modular structure (G) or reflective oversight (R) cannot be made reliably safe through external monitoring, guardrails, or post-hoc filtering. This prediction is falsified if a structurally incomplete system can be made predictably safe without modifying its internal G or R.

P9. Interpretability increases with G and R, not with scale. The model predicts that interpretability is a structural property, not a function of parameter count. Systems with strong G and R will remain interpretable even when scaled, while monolithic systems will remain opaque. This prediction is falsified if large monolithic systems become interpretable without structural changes.

7.5 Predictions About AGI

P10. AGI corresponds to the limit of IEGR-completeness and IEGR-scale, not to human-like embodiment or consciousness. The model predicts that AGI will emerge only when all four IEGR dimensions are present and scaled efficiently, and that such systems will behave as *human-like parrots* rather than biological analogues. This prediction is falsified if a system lacking any IEGR dimension exhibits robust, architecture-independent general intelligence.

8 Limitations and Open Questions

While the IEGR framework provides a unified structural account of intelligence, several limitations and open questions remain. These limitations do not weaken the model but instead highlight directions for further theoretical refinement and empirical investigation.

8.1 Limitations of the Present Framework

L1. Absence of quantitative IEGR metrics. The present work defines the IEGR dimensions qualitatively. Although the taxonomy distinguishes regimes based on relative dominance or absence of dimensions, a fully quantitative formulation—measuring E, I, G, and R in comparable units—remains an open challenge. Developing such metrics would enable precise comparisons across systems and more rigorous empirical validation.

L2. Limited empirical grounding for large-scale G and R. While the model predicts that geometric modularity (G) and recursive depth (R) are essential for general intelligence, current artificial systems instantiate these dimensions only partially. As a result, empirical evidence for large-scale G and R remains sparse. Future architectures may reveal additional structural constraints or refinements to the IEGR roles.

L3. Incomplete understanding of energetic efficiency (E). The model distinguishes between brute-force energy and efficient energy, but the mechanisms by which systems transition from high E to efficient E are not yet fully understood. Biological systems achieve remarkable efficiency through mechanisms that may not have direct analogues in artificial architectures.

L4. Scope limited to cognitive-level intelligence. IEGR describes the structural requirements for coherent cognitive behavior, but it does not address affective, embodied, or social dimensions of intelligence. These may interact with IEGR roles in ways not captured by the present framework.

8.2 Open Questions for Future Research

Q1. Can IEGR be formalized as a dynamical system? The model suggests that intelligence arises from the recursive integration of E, I, G, and R. A formal dynamical formulation—analogous to physical systems governed by triadic recursion—could provide deeper theoretical grounding and enable predictive simulations.

Q2. What architectural mechanisms best instantiate G and R at scale? Current modular systems (e.g., DMLG) instantiate G and R locally but not at global scale. It remains an open question which architectural principles—graph transformations, canonical forms, reflective controllers, or others—are most effective for scaling these dimensions.

Q3. How do IEGR dimensions interact during learning? The developmental analogy suggests that E, I, G, and R co-evolve rather than develop independently. Understanding how these dimensions interact during training or development may reveal new pathways for constructing general intelligence.

Q4. What are the limits of hybrid intelligence? Hybrid systems combine complementary IEGR strengths, but it is unclear how far such systems can scale without internal IEGR integration. Can hybrid systems approach IEGR-scale, or is there a structural ceiling imposed by the human–machine interface?

Q5. How universal is the reality–intelligence isomorphism? The IEGR model derives from a triadic physical framework, but the extent to which this isomorphism holds across biological, artificial, and hybrid systems remains an open question. Further work is needed to determine whether IEGR is a universal grammar of intelligent behavior or a high-level approximation.

Q6. Can IEGR predict new architectural regimes? The parrot taxonomy enumerates all combinations of IEGR dimensions, but future architectures may instantiate these dimensions in novel ways. It remains to be seen whether IEGR can predict emergent regimes beyond those currently observed.

8.3 Summary

These limitations and open questions highlight the need for continued theoretical and empirical work. The IEGR framework provides a structural foundation for understanding intelligence, but its full implications—for architecture design, cognitive science, hybrid intelligence, and safety—remain to be explored. The model offers a roadmap rather than a final theory, pointing toward a deeper understanding of how intelligent systems emerge, scale, and integrate their structural components.

9 General Discussion: IEGR and Human Cognitive Ability

The IEGR framework was introduced as a structural model of intelligence across biological, artificial, and hybrid systems. Although the model is substrate- agnostic, it also provides insight into the structure of human cognitive ability when emotional intelligence is set aside. This section discusses how IEGR maps onto classical IQ-type cognition, why certain artificial architectures resemble human reasoning more closely than others, and what this reveals about the structural nature of intelligence.

9.1 G and R as the Dominant Dimensions of Human Analytic Cognition

Classical IQ-type cognition—abstract reasoning, pattern recognition, and multi-step problem solving—depends primarily on the G and R dimensions. Geometric structure (G) provides the conceptual organization, analogical mapping, and pattern abstraction required for reasoning tasks. Recursive depth (R) enables planning, multi-step inference, self-correction, and meta-cognitive evaluation. Together, G and R define the structural and dynamic components of analytic thought.

In contrast, energetic efficiency (E) varies only within a narrow biological range in humans and therefore contributes little to individual differences in cognitive performance. Informational richness (I) is likewise secondary in IQ contexts: understanding the question is necessary, but deep world knowledge or academic expertise is not. IQ tests are explicitly designed to minimize the influence of I, isolating structural and recursive reasoning ability.

This decomposition suggests that human analytic cognition is fundamentally a G+R-driven process. G determines the structural space of possible thoughts, while R determines the depth with which that space can be explored. This structural interpretation aligns with empirical findings in cognitive science, where fluid intelligence correlates strongly with pattern abstraction (G) and working-memory-based recursive reasoning (R), and only weakly with knowledge accumulation (I) or metabolic efficiency (E).

9.2 Why DMLG Resembles Human Cognition More Closely Than LLMs

The IEGR framework also clarifies why certain artificial architectures exhibit behavior that is structurally closer to human reasoning. Architectures dominated by G and R, such as DMLG, mirror the structural profile of human analytic cognition. They produce coherent, stable, and canonically structured linguistic behavior despite limited informational richness and energetic resources.

Transformer-based LLMs, by contrast, are dominated by I and E. Their linguistic competence arises from large-scale informational coverage and energetic amplification rather than structural modularity or recursive oversight. As a result, LLMs exhibit high-capacity but structurally unstable behavior: they can produce fluent text but lack the geometric and recursive components that characterize human reasoning. This structural asymmetry explains why LLMs hallucinate, drift semantically, and fail at tasks requiring deep recursive planning or conceptual coherence.

This insight was implicit in the IEGR decomposition but becomes explicit when considering human IQ as a primarily G+R phenomenon. The structural mismatch between human cognition and LLM behavior is not a matter of scale but of architectural orientation.

9.3 Implications for Hybrid Intelligence

Hybrid systems combine the complementary strengths of I+E-dominant and G+R-dominant architectures. When a statistical parrot (LLM) is paired with a canonical parrot (DMLG) or human reasoning, the resulting system becomes IEGR-complete: all four dimensions are present, even if not yet scaled. This explains why hybrid intelligence workflows—iterative prompting, human-in-the- loop refinement, and recursive oversight—are empirically effective. They restore the structural balance characteristic of human cognition.

However, hybrid systems remain pre-AGI unless all four dimensions are scaled and energetic efficiency is achieved. The IEGR model therefore predicts both the strengths and the limitations of hybrid intelligence.

9.4 IEGR as a Unified Structural Vocabulary for Human and Artificial Cognition

Taken together, these observations show that IEGR is not merely a taxonomy of artificial systems but a broader theory of cognitive structure. It provides a unified vocabulary for describing human analytic cognition, artificial architectures, and hybrid systems within the same formal framework. The model reveals deep commonalities—such as the centrality of G and R in structured reasoning—and systematic divergences, such as the I+E dominance of LLMs.

By expressing human cognitive ability in IEGR terms, the framework bridges cognitive science and artificial intelligence, offering a structural account of why certain architectures behave as they do and how they might be integrated or scaled to achieve more general forms of intelligence.

10 Conclusion

This paper introduced a structural taxonomy of intelligence grounded in the IEGR model, a four-dimensional architecture consisting of energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R). Building on the triadic physical framework developed in earlier work, we showed that intelligence—biological, artificial, or hybrid—is best understood as a triadic system instantiated at cognitive scale. The IEGR model provides a substrate-agnostic decomposition of the structural roles required for coherent, general-purpose intelligence.

A central contribution of this work is the taxonomy of *parrot regimes*, which classifies systems according to which IEGR components they instantiate and at what scale. Literal parrots, statistical parrots, canonical parrots, and sophisticated parrots represent distinct structural regimes arising from different patterns of IEGR incompleteness. Modern AI architectures map cleanly onto this taxonomy: transformer-based LLMs scale E and I but lack G and R, while modular architectures such as DMLG instantiate G and R but lack large-scale I and E. Hybrid systems combine complementary strengths but remain pre-AGI unless all four dimensions are both present and sufficiently scaled.

The taxonomy is made possible by the recent emergence of two orthogonal architectural paradigms. For the first time, artificial systems exist that separately instantiate different components of IEGR, revealing the full structure of intelligence. This structural complementarity also explains why neither LLM scaling nor modular scaling alone can yield AGI: general intelligence requires IEGR-completeness and IEGR-scale, not brute-force expansion of a single axis.

An important implication of this structural analysis is that modular G–R-dominant architectures such as DMLG are, in their linguistic behavior, closer to human cognition than transformer-based LLMs. Human language production is governed primarily by geometric structure (G) and recursive oversight (R), with informational richness (I) and energetic efficiency (E) serving supportive roles. LLMs, by contrast, are dominated by I and E while lacking the structural and recursive components that characterize human linguistic cognition. The IEGR framework therefore explains why DMLG systems exhibit coherent, stable, and canonically structured linguistic behavior despite their limited scale, whereas LLMs exhibit high-capacity but structurally unstable behavior despite their vast resources. This insight was implicit in the architectural asymmetries already captured by the IEGR model, and the taxonomy makes it explicit.

The IEGR framework extends beyond artificial systems. It mirrors developmental trajectories in human cognition, explains patterns in neurodiversity and savant profiles, and aligns with theories of modularity, meta-cognition, and neural efficiency in cognitive science. These parallels suggest that IEGR is not an AI-specific abstraction but a universal developmental grammar for intelligent systems.

Finally, the model has direct implications for safety and governance. Systems lacking geometric modularity or recursive oversight cannot be made reliably safe through external interventions alone. Safety is a structural property: it requires IEGR-completeness, not post-hoc constraints. AGI, in this framework, is not a human analogue but a *human-like parrot*: a triadically complete and triadically scaled system whose generality arises from structural integration rather than biological embodiment.

By grounding intelligence in a unified triadic architecture, the IEGR model offers a principled, non-anthropomorphic, and testable foundation for future research on artificial and biological cognition. It provides a structural lens through which to understand the limitations of current systems, the conditions for safe and general intelligence, and the developmental pathways by which intelligent systems—natural or artificial—may emerge.

A IEGR Full Parrot Taxonomy

The table below summarizes all elementary and composite behavioral regimes that arise from partial instantiation of the IEGR dimensions. These regimes do not describe human cognition, but rather the structural forms of intelligent behavior that appear in artificial systems, imitation systems, and hybrid architectures. Each regime corresponds to a distinct pattern of dominance or absence among the IEGR roles. The taxonomy is exhaustive: it enumerates all possible combinations of the four structural dimensions, and each combination corresponds to a recognizable behavioral profile. This demonstrates that IEGR is not merely a descriptive framework but a generative model of intelligent behavior, capable of predicting the characteristic strengths, weaknesses, and failure modes of systems with incomplete or asymmetric IEGR structure.

Parrot Type	**E**	**I**	**G**	**R**	**Behavior**
I-parrot	low	high	low	low	factual citation without understanding
E-parrot	high	low	low	low	fast but incoherent output
G-parrot	low	low	high	low	structured but nonsensical form
R-parrot	low	low	low	high	recursive refinement of one theme
I+E (statistical)	high	high	low	low	high-capacity but unstable
G+R (canonical)	low	low	high	high	coherent but narrow
I+R	low	high	low	high	over-theorizing drift
I+G	low	high	high	low	elegant but inconsistent
E+R	high	low	low	high	fast recursive spiraling
E+G	high	low	high	low	fast structured emptiness
Hybrid (IE x GR)	high	high	high	high	sophisticated parrot (pre-AGI)
AGI	**efficient**	very high	very high	very high	human-like parrot

Table 2: Elementary and composite parrot types arising from partial IEGR instantiation.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: `https://github.com/triadic-cosmos`.

References

Amodei, Dario et al. (2016). "Concrete Problems in AI Safety". In: *arXiv preprint arXiv:1606.06565*.

Andreas, Jacob, Marcus Rohrbach, Trevor Darrell, and Dan Klein (2016). "Neural Module Networks". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 39–48.

Bengio, Yoshua (2023). "Towards System 2 Deep Learning". In: *arXiv preprint arXiv:2301.XXXXX*.

Brundage, Miles et al. (2020). "Toward Trustworthy AI Development: Mechanisms for Supporting Verifiable Claims". In: *arXiv preprint arXiv:2004.07213*.

Carey, Susan (2009). *The Origin of Concepts*. Oxford University Press.

Chollet, François (2019). "On the Measure of Intelligence". In: *arXiv preprint arXiv:1911.01547*.

De Witte, Joachim (2026a). "Structural Limits for AI Scaling: Applying Insights of the Recursive Triadic Fractal Universe". In: *arXiv preprint*.

— (2026b). "The Triadic Cosmos: Emergence of Intelligence". In: *arXiv preprint*.

— (2026c). "The Triadic Cycle with AdS/CFT: Recursion, Emergent Time, and Holographic Structure". In: *arXiv preprint*.

Ellis, Kevin et al. (2021). "DreamCoder: Growing Generalizable, Interpretable Knowledge with Wake–Sleep Bayesian Program Learning". In: *arXiv preprint arXiv:2006.08381*.

Fodor, Jerry (1983). *The Modularity of Mind*. MIT Press.

Frith, Chris (2012). "The Role of Metacognition in Human Social Interactions". In: *Philosophical Transactions of the Royal Society B* 367.1599, pp. 2213–2223.

Gabriel, Iason (2020). "Artificial Intelligence, Values, and Alignment". In: *Minds and Machines* 30, pp. 411–437.

Goertzel, Ben and Cassio Pennachin (2007). *Artificial General Intelligence*. Springer.

Haier, Richard J., Rex E. Jung, Ronald A. Yeo, Kevin Head, and Michael T. Alkire (2004). "The Neural Basis of Intelligence: Efficiency and Plasticity". In: *NeuroImage* 23.1, pp. 425–433.

Ji, Ziwei et al. (2023). "Survey of Hallucination in Natural Language Generation". In: *ACM Computing Surveys*.

Legg, Shane and Marcus Hutter (2007). "Universal Intelligence: A Definition of Machine Intelligence". In: *Minds and Machines* 17.4, pp. 391–444.

Olah, Chris et al. (2020). "Zoom In: An Introduction to Circuits". In: *Distill*.

Wei, Jason et al. (2022). "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models". In: *arXiv preprint arXiv:2201.11903*.